BIG JOE GOLD

GOLD TRADING & INVESTMENT

COMPLETE PROJECT AND OPERATING GUIDE

One workspace for gold trading operations.

A practical guide to customers, purchases, sales, stock, gold pricing, financing, finance ledgers, reporting, staff access, and the technology behind the application.

From a recorded purchase to a reconciled report

This guide explains what each workspace does, how records connect, what the figures mean, and which safeguards protect business records.

Audience
Owners, managers, operators, finance staff, and auditors.

App snapshot
Screens captured from the local application on 1 November 2026.

Reporting currency
GHS; gold weights appear in grams and kilograms, with pounds used on selected screens.

Prepared as a user-facing guide to the current Big Joe Gold project. Live prices, balances, permissions, and screens can change after capture.

START HERE

Contents & the big picture

The application is a local-first, database-backed workspace for a gold trading business. Its main purpose is to keep purchases, physical stock, sales, customer money, account movements, and reporting connected instead of relying on separate spreadsheets.

01 How the application fits together

02 Navigation, dashboard, and notifications

03 Customers, suppliers, and board prices

04 Receiving / buying gold

05 Sales, payments, returns, and reversals

06 Inventory, stock movements, and reconciliation

07 Finance accounts and payment workflows

08 Manual Bank & MOMO and financing

09 Reports overview: detailed metric guide

10 All report sections and exports

11 Staff access, audit, and SMS

12 Technology, setup, and safe operation

13 Quick troubleshooting and glossary

The core record flow

Customer or supplier › **Purchase or sale** › **Stock lot and movement** › **Payment and account** › **Reports and audit**

One important distinction

A **posted trade** is a real business transaction. A **draft** is work in progress. A **manual Bank & MOMO entry** is a separate reference ledger entry and does not create or alter a trade or system payment.

Record dates carefully

Trade date, payment date, movement date, and board-price effective time can differ. Reports filter by the date appropriate to each record, so correct dates are essential for matching statements.

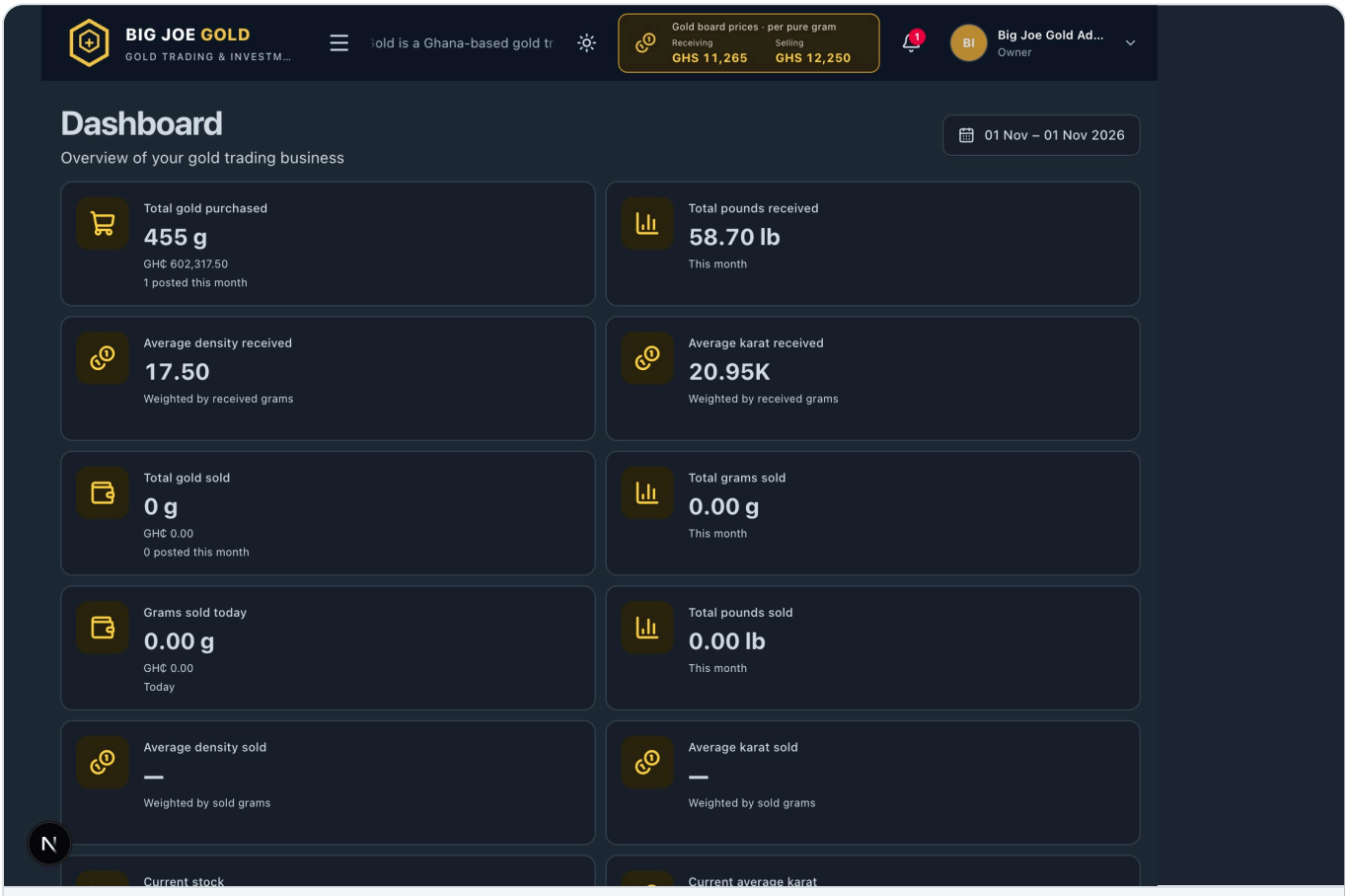
What is included

Trading and customer records; assay-based receiving; sales and stock allocation; FIFO inventory valuation; board-price history; customer advances and buyer financing; cash, bank, and Mobile Money accounts; manual ledgers; approvals; reporting; receipts; role-based access; and audit history.

Navigation, dashboard, and alerts

App frame

- **Side navigation:** opens Dashboard, Customers, Receive Stock, Sales, Add Stock, Inventory, Transactions, Gold Price, Financing, Manual Bank & MOMO, Finance, Reports, Staff Management, Audit Logs, and Settings. Visible destinations depend on access.
- **Top bar:** provides navigation controls, the theme toggle, current receiving/selling board rates, notifications, and the signed-in user menu.
- **Refresh:** reloads the screen's data. A refresh is useful after a transaction, but it does not fix a failed save; read the error message first.
- **Dates:** dashboard and report date filters define a period. Confirm the selected range before comparing totals.



Screenshot 1. Dashboard snapshot. Metrics shown are captured application data, not a forecast or a permanent balance.

1 Reporting period
Cards use the selected period or the stated today/month window.

2 Purchase and sales measures
Weight and GHS value are separate. A value of zero means no qualifying posted activity in that window.

3 Assay averages
Average density and karat are weighted by received/sold grams, not simple averages of transaction rows.

4 Lower dashboard panels
Scroll for current stock, price history, recent trades, outstanding balances, stock-by-karat, and activity charts.

> **Empty or dash values:** zero means no amount or weight in scope; a dash usually means a weighted average is unavailable because there are no qualifying grams. It is not necessarily an error.

Customers, suppliers, and gold board prices

Customers

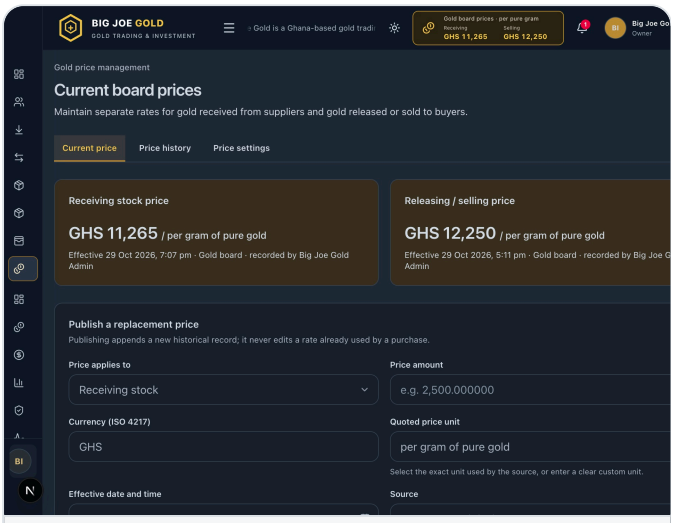
Customer records are separated by business role: sales buyers and receiving-stock suppliers. Profiles can include a generated customer code, individual/business type, phone and optional alternate phone, address, notes, active/archive status, and transaction history.

- Search and filter the list; use the profile to review the customer's linked trades and payments.
- Possible duplicate review helps operators avoid creating the same person twice.
- Archive is intended to retire a record from active use without erasing its history.
- CSV customer export is permission-gated; personal and identity information is restricted to authorized roles.

Board prices

Keep a separate current rate for **receiving stock** and for **releasing/selling gold**. Each published rate records currency, quoted unit, effective date/time, source, note, and publisher.

Publishing appends history; it does not rewrite a rate already recorded on a purchase. Confirm the correct direction, currency, and unit before publishing. The top-bar widget shows the current direction-specific prices.



Screenshot 2. Current board prices and the rate publication form.

1 Two directions
Receiving and selling rates are not interchangeable.

2 Unit matters
Check the quoted unit and currency as well as the number.

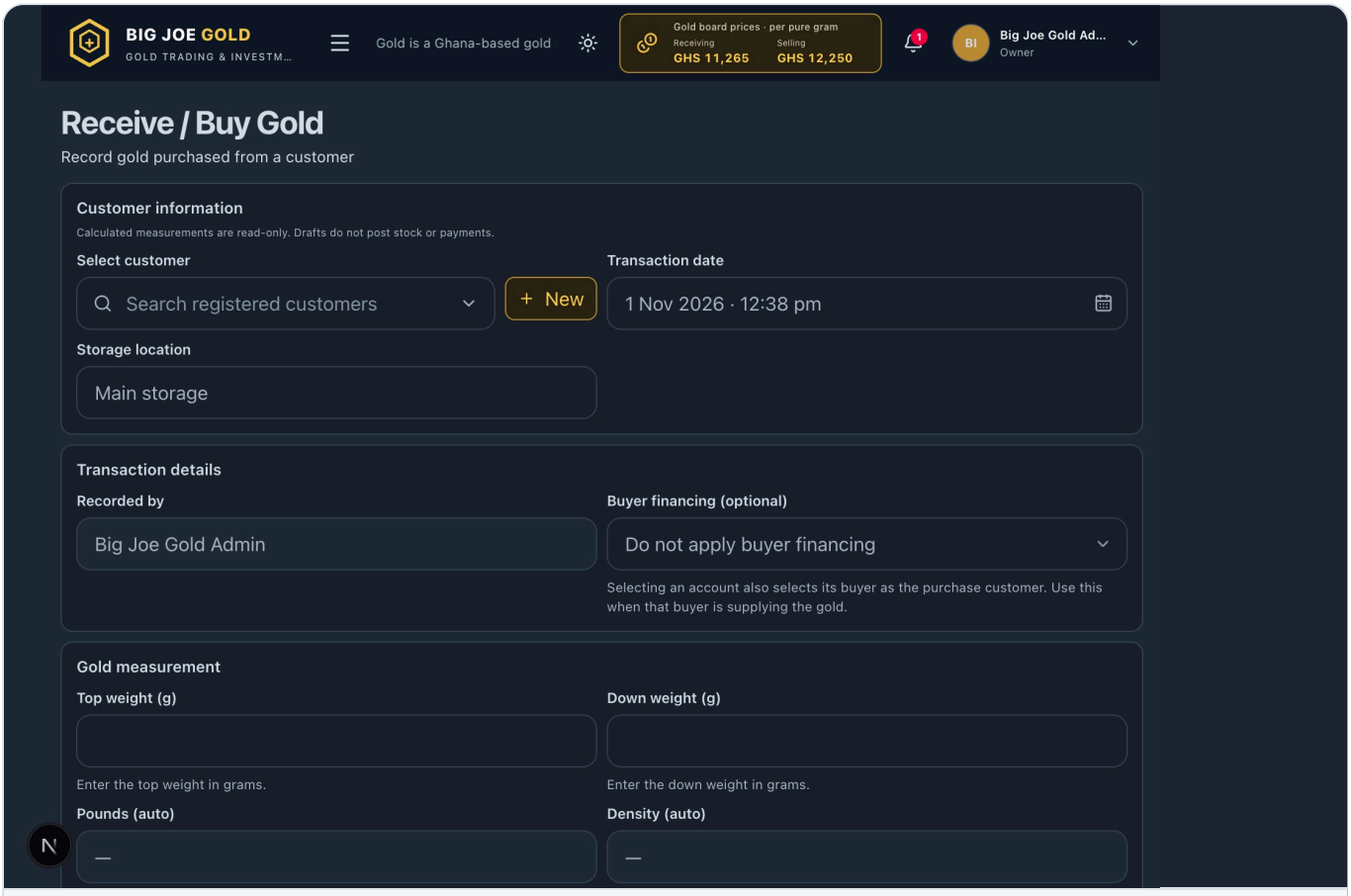
3 History is retained
Past transaction calculations keep their recorded price snapshot.

4 Settings
Price settings and receiving formulas are managed separately.

> **Formula governance:** Settings contains the server-side receiving calculation factors and required board-price unit. Changing a factor removes the prior confirmation. An authorized administrator must review and confirm the rules before a purchase can be completed; drafts remain available while rules are unconfirmed.

Receive / Buy Gold

This dedicated flow records gold bought from a customer/supplier. It connects the customer, date, location, assay inputs, calculated amount, optional payment, inventory lot, and receipt.



Screenshot 3. Receiving form. Calculated fields are read-only; the pictured form is blank for illustration.

1 Customer and date
Select a registered supplier/customer or create a record with the New action. Set the transaction date/time.

2 Location and financing
Choose where gold is stored. Apply buyer financing only when the supplying buyer has a matching account.

3 Assay inputs
Enter the measured top and down weights in grams. Derived pounds, density, karat, and purity are read-only.

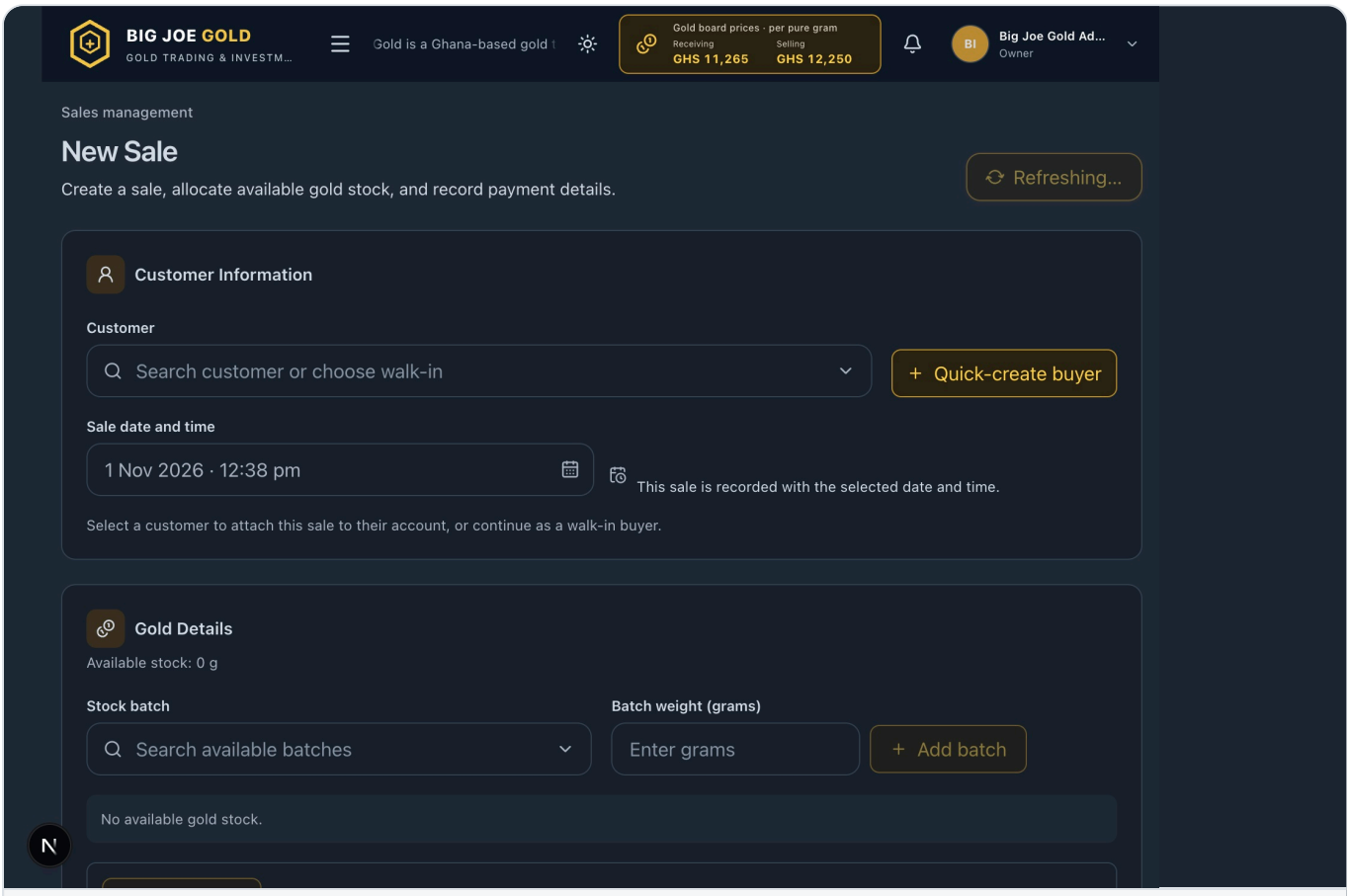
4 Payment and completion
Enter any amount paid and its payment details. Save a draft to check later, or complete only after reviewing the calculated purchase and confirmation.

What completion does

A draft does **not** add stock or post payments. Completing a purchase creates the posted trade, purchase measurement snapshot, inventory lot, stock receipt movement, any outgoing payment, and audit record together. If the write fails, those linked records are not meant to be partially created.

1. Choose supplier/customer, date, location, and any applicable buyer-finance account.
2. Enter the actual assay weights and review the calculated measurements and purchase value.
3. Record the amount paid now, account, method, and any reference; leave the balance outstanding if not fully paid.
4. Save a draft if details need checking; complete the purchase when ready to post.
5. Use the generated receipt to review or print the completed purchase.

Sales, payments, and corrections



Screenshot 4. New Sale screen. In this captured database snapshot, no gold was available for sale; a sale requires available stock.

1 Buyer
Choose a registered buyer, quick-create one if permitted, or record a walk-in buyer.

2 Batch allocation
Select available stock lot(s) and grams to release. The system prevents over-allocation or selling more stock than is available.

3 Price and payment
Review the calculated sale value, any permitted discount, amount received, account, and payment method.

4 Post and receipt
A posted sale issues stock from selected lots, saves the cost basis, records any payment, and makes a printable receipt available.

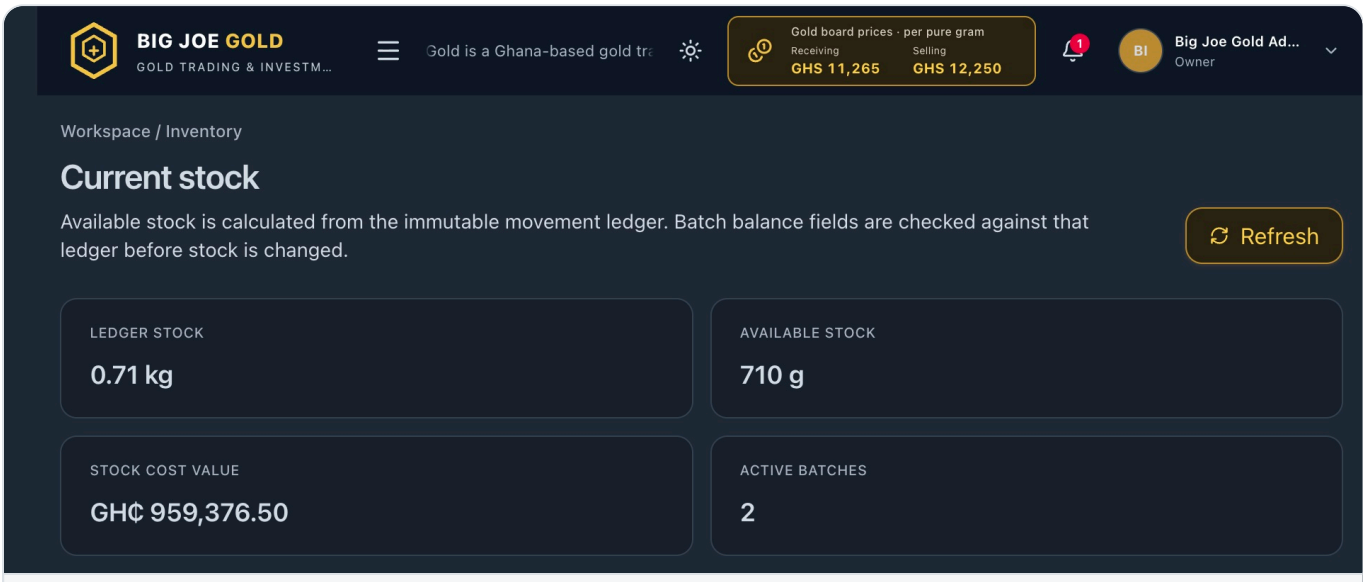
Sales history, payments, and returns

Sales screens separate new sale entry, all sales/history, sales payments, and returns/reversals. A payment must be allocated to the correct trade and direction: customer receipts are inbound; supplier payments are outbound. The server validates the allocation and outstanding balance.

Corrections use the dedicated return/reversal workflow and require an explanation. Stock returned to inventory is tied to the original sale and source lots. A full reversal is not a casual delete: follow the interface's return/reversal rules and review the audit trail.

FIFO and cost of sales
Inventory is consumed oldest available lot first (FIFO). The system stores each sale's lot allocations and cost basis so profit reflects the cost of the gold sold, not the current value of unsold stock.

Stock, movements, and reconciliation



Screenshot 5. Current stock summary. The batch table is intentionally omitted from this screenshot.

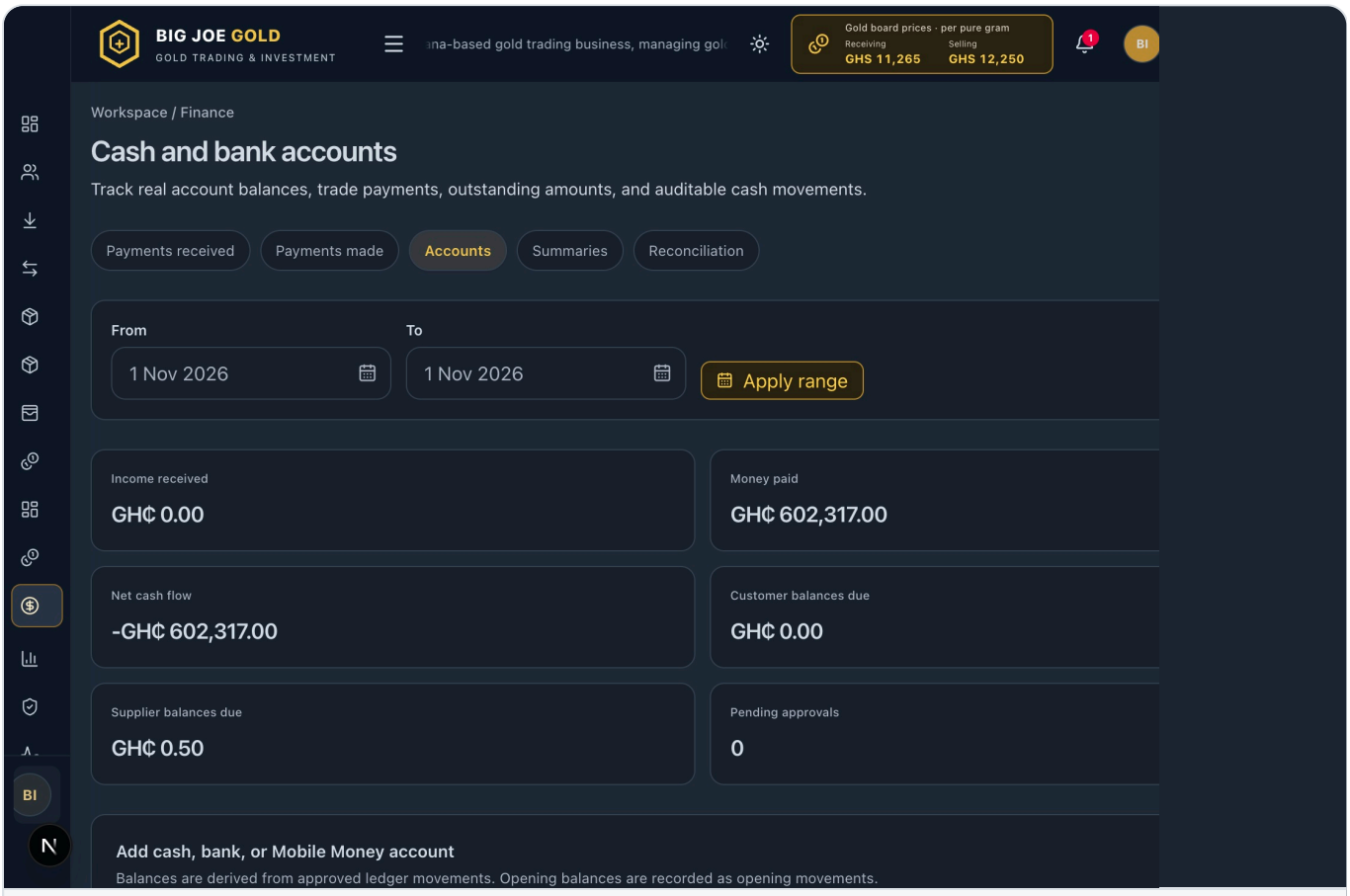
1. **Ledger stock**
   The calculated balance from posted movement records.
2. **Available stock**
   Gold that can be allocated to a new sale.
3. **Stock cost value**
   Recorded cost value of current stock; not a current market quote.
4. **Active batches**
   Stock is tracked by lot and location, not only as one total.

Inventory workspaces

- **Current Stock / Gold Batches:** inspect remaining grams, received weight, purity, density, location, and recorded cost for each lot.
- **Opening Stock:** establish an initial quantity with a date/reference; this creates a stock movement.
- **Manual Adjustment:** correct a real stock discrepancy with a reason and authorized permission.
- **Stock Transfers:** move quantity between storage locations; linked out/in movements preserve the total.
- **Stock Movements:** audit receipts, sale issues, returns, adjustments, and transfers in date order.
- **Alerts / Reconciliation:** investigate ledger mismatches, negative quantities, and batches needing attention before further stock changes.

> **Reconcile before adjusting.** Stock is derived from immutable movement history. If the displayed batch fields do not agree with the ledger, correct the underlying discrepancy through the reconciliation workflow rather than bypassing the ledger.

System payments and real account balances



Screenshot 6. Cash and bank account workspace. Blank fields are illustrative; existing account details are not shown.

1. **Finance tabs**
   Payments received, payments made, accounts, summaries, and reconciliation are distinct views.
2. **Date range**
   Summary and movement views can be filtered to a period.
3. **Accounts**
   Create cash, bank, or Mobile Money accounts with currency, provider, optional reference, and opening balance.
4. **Ledger-based balances**
   Balances derive from approved cash movements; opening balance is recorded as a movement.

Payment directions

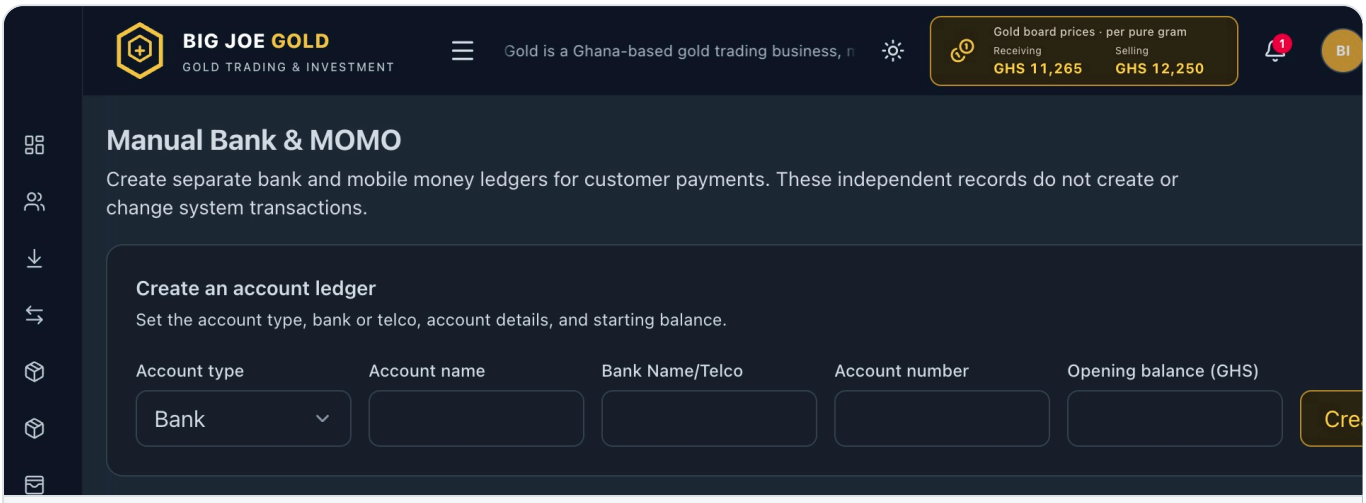
Payments received
Money received from customers; normally inbound and allocated to a posted sale, financing account, or supported customer transaction.

Payments made
Money paid out, including payments allocated to supplier purchase trades. The payment account and method are stored with the record.

Use account movements for transfers and adjustments, then reconcile unassigned or unmatched payment records to the correct cash/bank account. Where thresholds and roles require it, high-value adjustments wait for approval. Rejected or pending movements are not treated as available balance in the same way as approved movements.

> **Keep ledgers distinct:** A system payment updates the trade/payment record and account ledger. The Manual Bank & MOMO screen is separate and does not automatically reconcile with this finance ledger.

Manual Bank & MOMO and financing



Screenshot 7. Manual ledger setup fields. Account-holder details and existing ledgers are excluded from this image.

1. **Account type and name**
   Choose Bank or Mobile Money and a clear internal account name.
2. **Provider details**
   Record Bank Name/Telco and account number so the manual ledger is identifiable.
3. **Starting amount**
   Enter an opening balance; each ledger keeps its own opening balance and running statement.
4. **Record money flows**
   Received means customer money paid to you; paid means money you sent to a customer.

Each manual entry can retain customer name/phone, transaction reference, date, direction, note, and amount. The statement shows opening balance, period receipts, period payments, and closing/running balance. These records are for independent reconciliation/reference only—they do not create an official Trade or Payment.

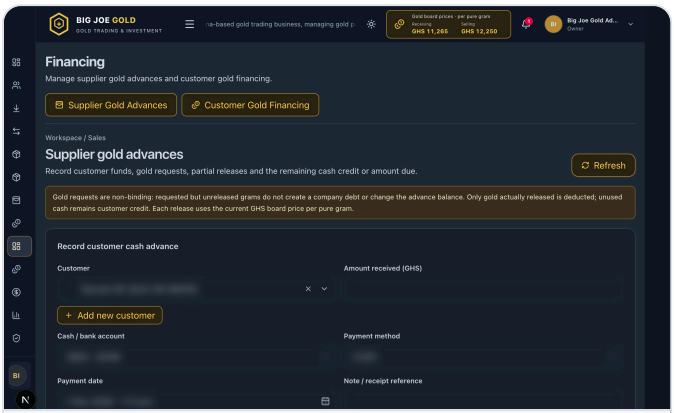
Financing tools

Supplier Gold Advances
Record cash received from a customer, track requests for gold, and record actual partial/full gold releases. Requested but unreleased grams do not create a company debt or change the balance. Only released gold is valued at the current compatible selling board price; unused cash remains customer credit.

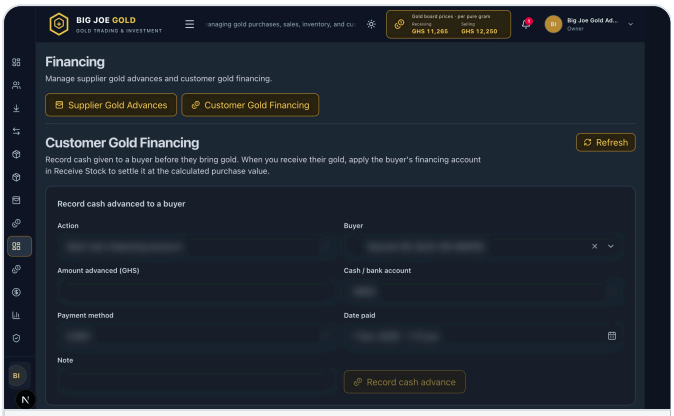
Customer Gold Financing
Record cash advanced to a buyer before they bring gold. Later, in Receive Stock, select that buyer's financing account to apply the purchase value against the financing position. Account activity links advances and settlement receipts.

Supplier advances and customer gold financing

The Financing workspace has two distinct tabs. Supplier Gold Advances tracks customer cash held for future gold releases; Customer Gold Financing tracks cash advanced to a buyer before they supply gold.



1 · Supplier Gold Advances. Record customer funds, request gold, and release measured gold from available stock. Personal and account values are obscured.



2 · Customer Gold Financing. Record cash advanced to a buyer and select its cash account, payment method, date, and note. Personal and account values are obscured.

1 Record a supplier advance

Choose the customer, cash/bank account, amount, payment method, and date. The receipt and account history document the funds received.

2 Request is not a release

A gold request is non-binding: requested but unreleased grams do not create a company debt or change the advance balance. Unused cash remains customer credit.

3 Measure and fulfill

Select an open request and available inventory batch, enter the actual assay measurements, and confirm the calculated value against the current compatible GHS board price. Only gold actually released is deducted.

4 Settle buyer financing through receiving

Record the cash advance to a buyer. Later, in Receive Stock, select that buyer's financing account when recording their gold; the purchase value is applied against the financing position.

Balances and history

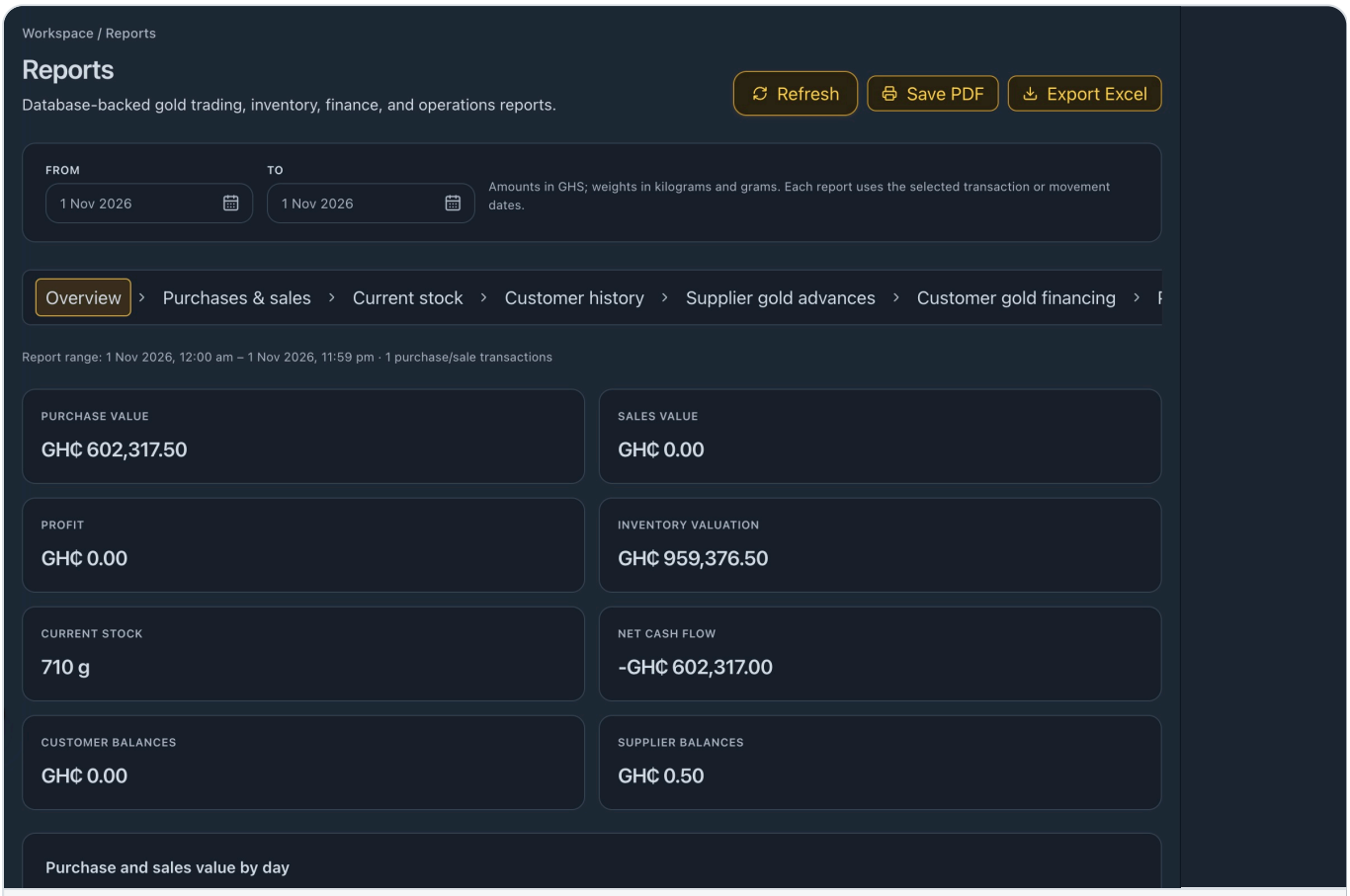
Supplier advance activity shows cash received, additional funds or refunds, requested and fulfilled gold, and remaining customer credit or amount due. Buyer financing shows cash advanced, gold value applied, and whether the customer owes the business or the business owes the customer.

Keep the ledgers distinct

Financing creates linked transaction and payment records. It is not the independent Manual Bank & MOMO reference ledger. Choose the correct cash/bank account and verify the receipt and account history after each posted action.

Before releasing gold: Confirm the selected customer request, available batch and grams, measured weights, purity, current GHS price, and calculated value. Requested grams alone do not reduce the advance.

Understand the report overview



Screenshot 8. Reports overview snapshot from 1 November 2026. Amounts and balances are illustrative of that selected date, not guaranteed current totals.

1 Date range
The displayed period here is 1 Nov 2026. Most activity totals follow transaction dates; other tabs use their own movement or statement dates.

2 Purchase and sales value
Value of qualifying purchase/sale trades in the selected period, not cash paid/received.

3 Profit and inventory value
Profit is on gold sold with a recorded cost basis. Inventory valuation is book cost of stock remaining.

4 Stock, cash flow, balances
Current stock and open balances describe positions; net cash flow describes money movement in the period.

The report screen provides a common From/To filter, refresh, browser-print **Save PDF**, and multi-sheet **Export Excel**. The selected dates must include the transaction/movement date you intend to examine.

> **Current example:** purchase value GH¢602,317.50, sales GH¢0.00, profit GH¢0.00, stock book value GH¢959,376.50, stock 710 g, net cash flow −GH¢602,317.00, customer balances GH¢0.00, supplier balances GH¢0.50. These values demonstrate how the tiles differ; they will change as records and dates change.

09 · METRIC DICTIONARY

Overview metrics, explained

Metric	Meaning and interpretation	Example
Purchase value	Total purchase-trade value recorded during the selected period. It is not necessarily the amount paid in cash during that same period.	GH¢602,317.50
Sales value	Net sales-trade value in the selected period, after applicable sale credits/corrections. Compare it with purchase value only with matching dates and scope.	GH¢0.00
Profit	Net sales value minus recorded cost basis of gold sold/released. It is not sales minus all purchases; unsold inventory is not counted as current-period profit. If any sale lacks a cost basis, profit may be unavailable rather than a reliable zero.	GH¢0.00
Inventory valuation	FIFO book value of remaining inventory based on saved lot costs. It is not the current selling price, cash available, or total historical purchases.	GH¢959,376.50
Current stock	Physical weight currently on hand after receipts, sales, returns, adjustments, and transfers; this is a position, not a period flow.	710 g = 0.710 kg
Net cash flow	Cash received minus cash paid in the selected period. Negative means recorded money out exceeded money in; it does not by itself mean the business lost that amount.	−GH¢602,317.00
Customer balances	Outstanding amounts due to the business on open customer sales. A zero means no qualifying open receivable in scope.	GH¢0.00
Supplier balances	Outstanding amount still payable on open purchase trades. A small remainder can persist even when most of a purchase has been paid.	GH¢0.50

Why purchase value can differ from inventory value
Purchase value is a period activity measure. Inventory valuation is the cost of the quantity still held at report time. Some purchases may already have been sold, and current stock may include older purchases.

Why cash flow differs from profit
Cash flow follows receipts and payments; profit follows sales value and cost of gold sold. Timing, unpaid balances, customer advances, and unsold stock can make them very different.

Report values are calculated from stored business records and role permissions. If a number seems wrong, verify the date range, posting status, payment direction, trade allocation, inventory movement, and access scope before changing data.

Report sections and exports

Purchases & sales

Group purchase/sale weight and value by day, week, or month; inspect trade details for the selected range.

Current stock

Stock by karat and batch, including remaining weight, assay, location, and FIFO book value.

Customer history

Review a customer's activity and transaction history, subject to personal-data permissions.

Supplier gold advances

See advance receipt, requests/releases, and remaining customer credit or balance position.

Customer gold financing

Review cash advanced to buyers and purchase amounts applied to their accounts.

Payments & balances

Period payments and open customer/supplier trade balances.

Account statements

Account-level entries and balances for selected dates; use for reconciliation.

Cash flow

Money received, money paid, net cash flow, and account cash movements.

Stock movements

Receipts, sale issues, returns, transfers, and adjustments during the period.

Staff activity

Recorded trade, payment, stock, and board-price actions, subject to staff-name permissions.

Board price history

Effective rates, direction, currency, unit, source, and publisher over time.

Save or share a report

1. Select the From and To dates first; apply the range where the screen provides an apply button.
2. Choose the tab you need. Use report tables and filters rather than treating the overview as a detailed ledger.
3. Use **Save PDF** to open the browser print dialog; select Save as PDF and confirm the page range/orientation.
4. Use **Export Excel** for a workbook with multiple sheets. Check the exported period and workbook before sending it outside the business.

> **Permission-aware reports:** Customer names, staff names, and personal details may be masked or omitted when the current user lacks the specific permission. A "Restricted" value is an access boundary, not a database error.

11 · ACCOUNTABILITY

Staff roles, audit history, and notifications

Roles and permissions

Access is permission-based and checked by server APIs, not just by hiding buttons. Built-in role templates include Owner, Manager, Gold Receiver / Buyer, Sales Officer, Inventory Officer, Accountant, Auditor, and legacy Trader / Viewer roles. Owners can configure per-user access for purchases, sales, customers, inventory, payments, finance approvals, reports, board prices, staff, audit, and settings.

- Only an Owner can assign the Owner role; staff cannot change their own access.
- Personal customer details, customer exports, sensitive finance actions, reversals, price changes, and staff management can require separate permissions.
- High-value account adjustments may require an authorized approval. Pending/rejected requests should be reviewed in the relevant approval workflow.
- Permission changes apply on the next server request; after access changes, have the staff member sign in again if needed.

Audit log

Audit entries record who made important changes and when. Core trade, stock, and finance writes and their audit records are designed to commit together. Staff/access changes include before/after details with secret-like fields redacted. Review the audit log when investigating an unexpected balance or an action that needs attribution.

Sign-in and staff codes

Owners use the configured email/password account. Staff use the generated BJG-prefixed four-digit access code issued by an owner. Code generation replaces the previous code; viewing a code is audit logged. Keep access codes private and deactivate staff promptly when access should end.

Owner SMS

When Wigal FROG settings and an approved sender ID are configured, owner notifications may be sent for completed purchases, posted sales, and payments recorded from the Payments workspace. SMS is best-effort: the provider or mobile network can reject, delay, or fail delivery. SMS must not be used as proof that a transaction saved; verify the in-app receipt and ledger.

Operational rule
Do not share screenshots, exports, receipts, access codes, customer IDs, bank account numbers, or API credentials outside authorized channels. This guide’s screenshots omit personal account details.

Technology, setup, and data safeguards

What the project is built with

Application
Next.js App Router, React, TypeScript, Tailwind CSS, reusable UI components, and Recharts for data visualizations.

Data and validation
MySQL through Prisma; Zod validates request data; fixed-precision decimal types protect currency and measurement calculations.

Identity and access
Better Auth for identity/session handling, database-backed roles and permission grants, and rate limits on selected sensitive APIs.

Core ledgers
Trade, purchase measurement, inventory lot/movement, payment, cash account/movement, financing, sale correction, and audit models preserve linked business history.

Typical repository layout

`src/app/`	Pages, route handlers, authentication endpoints, and REST-style APIs.
`src/components/`	Dashboard and feature screens such as receiving, sales, finance, inventory, and reports.
`src/lib/`	Authorization, validation, business calculations, shared data access, SMS, and formatters.
`prisma/schema.prisma`	Relational data model; forward migrations live under `prisma/migrations/`.
`tests/`	Automated tests for formulas, validation, permissions, reports, and ledger calculations.

Local development and deployment

Requirements are Node.js 20+, npm, and MySQL 8+. Configure `DATABASE_URL`, `BETTER_AUTH_SECRET` (at least 32 characters), the authentication URL, the staff access-code encryption key, and initial owner values in a local `.env`. Never commit `.env` or use example credentials for real data.

Common commands: `npm install`, `npm run db:generate`, `npm run db:migrate`, `npm run db:seed`, `npm run dev`; use `npm run db:deploy` to apply reviewed migrations outside development. Keep encrypted access-code keys backed up; losing the key makes stored encrypted codes unreadable.

Data and business safeguards

- Core purchase, sale, stock, payment, and audit operations use atomic database transactions.
- Gold trade values use pure weight and the selected unit-compatible board rate; stored purchase values are snapshots and are not recalculated when the board changes.
- Purchases create lots; sales use FIFO lots. Insufficient stock blocks the sale instead of partially posting it.
- Back up MySQL, restrict database access, use TLS for deployed connections, secure the authentication secret, and test restore procedures.

Troubleshooting and glossary

What you see	What to check next
Purchase saved as draft; stock did not increase	This is expected. Review the measurements and formula confirmation, then complete/post the purchase.
Sales form has no available batches	Check current stock, batch status, location, and stock reconciliation. A draft purchase does not provide saleable stock.
A report shows zero	Confirm the date range and posting status. Verify whether the report uses trade date, paid date, or movement date.
Profit is unavailable or zero	Profit requires sales and a recorded cost basis. Zero sales can correctly produce zero profit; incomplete cost basis can make profit unavailable.
Bank/account balance looks wrong	Review the account's opening movement, payments assigned to the account, transfers, adjustment approvals, and reconciliation status.
Manual ledger disagrees with Finance	They are independent ledgers. Reconcile them manually; creating a manual entry does not update a system payment/account.
User cannot see a screen or customer details	Ask an Owner to check the user's role and specific permission. Sensitive data can be hidden even if a report is visible.
SMS not received	Check provider acceptance, approved sender ID, recipient number, and all required configuration. Verify the in-app transaction independently.
Save fails or loading continues	Read the app error, confirm required fields and permissions, refresh once, and verify whether a receipt/ledger record exists before retrying.

Glossary

g / kg
Gram / kilogram. 1 kg = 1,000 g.

lb
Pound. Some receiving/dashboard fields show pounds alongside grams.

Karat / purity
Purity measures the gold content; calculated karat and purity are saved with the purchase assay.

FIFO
First In, First Out: oldest available inventory lots are consumed first for cost-of-sales tracking.

Cash flow
Money received minus money paid over a period; separate from accounting profit.

Outstanding balance
Trade value not yet settled by correctly allocated payments or supported financing settlement.

Board price
Published market rate with direction, currency, unit, and effective time.

Movement ledger
Immutable event history used to derive current stock or account balances.

Closing principle: Record the real event once, choose the correct direction/date/account, verify the receipt or ledger, and use reports to review—not replace—the underlying transaction history.